

MAKESHKUMAR M

AI Engineer • Generative AI & Agentic Systems | Frontend (MERN) Foundation
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PROFILE

Frontend developer (React.js, Next.js) transitioning into AI Engineering, currently completing an intensive AI Engineer track covering LLM engineering, Retrieval-Augmented Generation (RAG), QLoRA fine-tuning, agentic systems, MCP, and MLOps for deploying Generative AI at scale. Building production-oriented AI projects — including a citation-backed, no-hallucination RAG platform for K-12 education — while applying strong full-stack fundamentals to ship end-to-end AI products, from data pipelines to deployed UI.

SKILLS

AI / ML Engineering: LLM Engineering, RAG Architecture, Prompt Engineering, QLoRA Fine-Tuning, Agentic AI, Model Context Protocol (MCP), Vector Search

AI Tooling & Infra: ChromaDB, LanceDB, Voyage AI Embeddings, LangChain-style orchestration, Bolt, Lovable, Cursor IDE

MLOps & Deployment: Deploying GenAI/Agentic apps at scale, API integration, Vercel, Git/GitHub, environment & pipeline management

Web Development: React.js, Next.js, JavaScript (ES6+), Node.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Framer Motion

Languages & Data: JavaScript, Python (learning), REST APIs, JSON, Excel / data handling

Tools: VS Code, Git, GitHub, Vercel, Postman

AI ENGINEER TRAINING (IN PROGRESS)

- **Core Track** — LLM Engineering, Retrieval-Augmented Generation (RAG), QLoRA fine-tuning, and building production-grade AI agents.
- **Agentic Track** — Designing and building autonomous agents and tool-use systems with the Model Context Protocol (MCP).
- **MLOps Track** — Deploying Generative AI and Agentic AI applications at scale, covering pipelines, monitoring, and infra for production AI systems.

AI PROJECTS

AI-Powered K-12 Educational RAG Platform | *RAG, ChromaDB / LanceDB, Voyage AI Embeddings, MCP*

- Architecting a Retrieval-Augmented Generation system to index Class 1–12 textbooks and serve accurate, citation-backed answers with a strict no-hallucination requirement.
- Evaluating ChromaDB vs. LanceDB for vector storage and Voyage AI for embeddings to optimize retrieval accuracy and latency.
- Planning MCP-based tool integration to extend the platform toward agentic, curriculum-aware study assistance.

Personal Knowledge Management “Second Brain” System | *Vector DB, Embeddings, Agentic Workflows*

- Designing a personal knowledge base as a foundation for a JARVIS-style agentic assistant, focused first on reliable retrieval before layering autonomous, tool-using agent capabilities.

AI & Labor Market Impact Report | *Data Research, Interactive HTML/JS*

- Produced a detailed analysis of global AI-driven job disruption through 2035, delivered as a rich interactive web report.

FinFlow — Personal Finance Dashboard | *React 18, Vite, Tailwind CSS, Recharts, Zustand*

- Built a full personal finance dashboard with role-based UI, light/dark mode, data visualizations, and CSV export as part of a frontend assessment.

EXPERIENCE

Web Developer Intern — *IEDEO* [Mar 2025 – Jun 2025]

- Developed and optimized responsive web applications using React.js and Next.js, ensuring cross-browser compatibility and performance.
- Leveraged AI-powered development tools (Bolt, Lovable, Cursor IDE) to streamline workflows and integrate AI-driven features into applications — early hands-on exposure to AI-assisted engineering.
- Built and deployed a dynamic landing page using Next.js, Tailwind CSS, and Framer Motion, improving responsiveness and interactivity.

Data Entry Operator — *SBL Knowledge Services Limited* [Oct 2024 – Dec 2024]

- Maintained large volumes of structured data with 100% accuracy using MS Excel and database tools, including cleaning and verification — foundational data-handling discipline relevant to dataset preparation for AI systems.

EDUCATION

B.E., Electronics and Communication Engineering — Kamaraj College of Engineering and Technology | 2016 – 2020